Lecture 5 Compendium: Aerial Robotics (UAVs) in Agriculture

Module 2: Technologies of IoT in Crop Production

Dr. Kadeghe Fue
PhD Agroecology

Compendium for Advanced Study

Contents

1 Introduction: The Low-Altitude Remote Sensing Revolution **2**

2 UAV Architectures & Aerodynamics **2**
2.1 Rotary-Wing (Multi-Rotor) Physics 2
2.2 Fixed-Wing Physics 2

3 The Payload: Sensors & Radiometry **3**
3.1 Multispectral Imaging 3
3.2 Thermal Imaging (Thermography) 3

4 Photogrammetry: The Mathematics of Mapping **3**
4.1 Ground Sampling Distance (GSD) 3
4.2 Image Footprint 3
4.3 Python Flight Planner 4

5 Operational Algorithms: Path Planning **5**
5.1 The Wind Triangle 5

6 Advanced Application: Aerial Spraying **5**
6.1 Droplet Physics 5

7 Regulations & Safety (Tanzania Context) **5**

8 Review Questions & Lab Assignment **6**

1 Introduction: The Low-Altitude Remote Sensing Revolution

Satellites (Macro-scale) suffer from cloud cover and low spatial resolution (10m to 30m). Ground sensors (Micro-scale) lack spatial coverage. **Unmanned Aerial Vehicles (UAVs)**, or drones, fill the critical **Meso-scale** gap (1cm to 5cm resolution).

In this PhD module, we move beyond "flying for fun." We treat the drone as a rigorous scientific instrument. We must understand:

1. **Aerodynamics:** How to keep the sensor in the air efficiently.
2. **Photogrammetry:** How to reconstruct 3D reality from 2D images.
3. **Radiometry:** How to measure light to infer plant health.

2 UAV Architectures & Aerodynamics

The choice of platform is not aesthetic; it is determined by the physics of the mission profile.

2.1 Rotary-Wing (Multi-Rotor) Physics

Quadcopters generate lift via **Momentum Theory**. The propeller pushes air down (F_{thrust}), creating an equal and opposite force upwards.

Power Consumption (P): For a rotor generating thrust T in hover:

$$P = \frac{T^{3/2}}{\sqrt{2\rho A}} \tag{1}$$

Where:

- ρ: Air density ($1.225\,\text{kg/m}^3$ at sea level).
- A: Disk area of the propeller.

Implication: Power consumption scales with $T^{3/2}$. Heavy payloads kill battery life exponentially. Small propellers (small A) are less efficient than large ones.

2.2 Fixed-Wing Physics

Fixed-wing drones generate lift using airfoils (Bernoulli's Principle). They are significantly more efficient because the motor only fights Drag, not Weight (Lift fights Weight).

Breguet Range Equation (Electric):

$$R = E^* \cdot \eta_{total} \cdot \frac{L}{D} \cdot \frac{m_{batt}}{m_{total}} \tag{2}$$

Where:

- R: Range.
- E^*: Specific energy of battery (Wh/kg).
- L/D: Lift-to-Drag ratio (Aerodynamic efficiency).
- η_{total}: Propulsive efficiency.

Comparison

Multi-Rotor: Low $L/D \approx 0$ (Requires constant power). Flight time: 20-30 mins.
Fixed-Wing: High $L/D \approx 10 - 15$. Flight time: 60-120 mins.

3 The Payload: Sensors & Radiometry

The drone is just the bus; the sensor is the passenger doing the work.

3.1 Multispectral Imaging

Plants interact with light in specific ways.

- **Visible Blue/Red:** Absorbed by Chlorophyll (Photosynthesis).
- **Visible Green:** Reflected (Why plants look green).
- **Near Infrared (NIR):** Strongly reflected by the spongy mesophyll structure of healthy leaves.

The Red Edge: The sharp transition region between Red absorption (≈ 670nm) and NIR reflectance (≈ 750nm). Stress shifts this curve.

3.2 Thermal Imaging (Thermography)

Used to measure **Canopy Temperature**. **Physics (Stefan-Boltzmann Law):**

$$j^* = \varepsilon \sigma T^4 \tag{3}$$

- **Application:** Stressed plants close stomata to conserve water $\rightarrow$ Transpiration stops $\rightarrow$ Evaporative cooling stops $\rightarrow$ Leaf temperature rises.
- **Correction:** Data must be corrected for Emissivity ($\varepsilon \approx 0.98$ for vegetation) and Background Temperature.

4 Photogrammetry: The Mathematics of Mapping

We use **Structure from Motion (SfM)** to stitch thousands of overlapping 2D images into a 3D Orthomosaic.

4.1 Ground Sampling Distance (GSD)

GSD is the most critical parameter. It represents the physical distance on the ground covered by one pixel.

$$GSD = \frac{S_w \cdot H}{F_r \cdot I_w} \tag{4}$$

Where:

- S_w: Sensor Width (mm).
- H: Flight Altitude (m).
- F_r: Focal Length (mm).
- I_w: Image Width (pixels).

4.2 Image Footprint

How much land does one photo cover?

$$D_w = \frac{S_w \cdot H}{F_r} \quad , \quad D_h = \frac{S_h \cdot H}{F_r} \tag{5}$$

4.3 Python Flight Planner

This script calculates the mission parameters required to achieve a target GSD and Overlap.

```python
import math

class DroneMission:
    def __init__(self, sensor_width_mm, focal_length_mm, image_width_px, image_height_px):
        self.Sw = sensor_width_mm
        self.Fr = focal_length_mm
        self.Iw = image_width_px
        self.Ih = image_height_px

    def calculate_gsd(self, altitude_m):
        """Calculates Ground Sampling Distance in cm/pixel"""
        gsd_cm = (self.Sw * altitude_m * 100) / (self.Fr * self.Iw)
        return gsd_cm

    def calculate_altitude(self, target_gsd_cm):
        """Calculates required altitude for a target GSD"""
        altitude = (target_gsd_cm * self.Fr * self.Iw) / (self.Sw * 100)
        return altitude

    def calculate_coverage(self, altitude_m, overlap_front_percent, overlap_side_percent):
        """Calculates image footprint and spacing between shots"""
        # Footprint on ground (meters)
        footprint_w = (self.Sw * altitude_m) / self.Fr
        footprint_h = (self.Sw * (self.Ih/self.Iw) * altitude_m) / self.Fr

        # Shooting Interval (Distance between photos)
        # Front overlap determines trigger distance
        dist_between_shots = footprint_h * (1 - overlap_front_percent/100)

        # Transect Spacing (Distance between flight lines)
        # Side overlap determines line spacing
        dist_between_lines = footprint_w * (1 - overlap_side_percent/100)

        return {
            "footprint_width_m": footprint_w,
            "footprint_height_m": footprint_h,
            "trigger_distance_m": dist_between_shots,
            "line_spacing_m": dist_between_lines
        }

# --- EXAMPLE: DJI Phantom 4 Pro ---
# Sensor: 1-inch (13.2mm width), Focal: 8.8mm, Res: 5472x3648
dji_p4p = DroneMission(13.2, 8.8, 5472, 3648)

# Target: 2 cm/pixel GSD for Weed Detection
target_gsd = 2.0
altitude = dji_p4p.calculate_altitude(target_gsd)

print(f"--- Mission Plan for GSD {target_gsd} cm/px ---")
print(f"Required Altitude: {altitude:.1f} meters")

# Planning Overlap
ops = dji_p4p.calculate_coverage(altitude, overlap_front_percent=80, overlap_side_percent=75)

print(f"Trigger Camera Every: {ops['trigger_distance_m']:.1f} meters")
print(f"Flight Line Spacing: {ops['line_spacing_m']:.1f} meters")
```

Listing 1: UAV Mission Planner Class

5 Operational Algorithms: Path Planning

To cover a field efficiently, we use the **Boustrophedon** (Ox-turning) path.

5.1 The Wind Triangle

Drones drift. The flight controller must compensate for wind vector $\vec{W}$ to maintain the ground track $\vec{T}$.

$$\vec{V}_{ground} = \vec{V}_{air} + \vec{W} \tag{6}$$

If flying cross-wind, the drone must "crab" (yaw into the wind). This reduces effective ground speed and battery range.

Research Tip: Always plan flight lines *perpendicular* to the wind direction for Fixed Wings (to minimize ground speed variation), but *parallel* for spraying runs to minimize drift.

6 Advanced Application: Aerial Spraying

Spraying drones (e.g., DJI Agras) are replacing knapsack sprayers. This is not just about flying; it is fluid dynamics.

6.1 Droplet Physics

We must avoid **Drift** (chemical blowing away). Drift risk is proportional to:

$$Risk \propto \frac{WindSpeed \cdot Altitude}{DropletSize^2} \tag{7}$$

- **ULV (Ultra Low Volume):** Drones use high concentration, low water volume.
- **Downwash:** The propellers create a vortex that pushes the canopy open, allowing droplets to penetrate to the lower leaves.

7 Regulations & Safety (Tanzania Context)

Agroecology PhDs must adhere to the **Tanzania Civil Aviation Authority (TCAA)** regulations.

1. **Visual Line of Sight (VLOS):** You must be able to see the drone with the naked eye (typically < 500m radius).
2. **Altitude Cap:** Max 120m (400ft) AGL (Above Ground Level) to avoid manned aircraft.
3. **Registration:** All commercial/research drones must be registered.
4. **BVLOS (Beyond Visual Line of Sight):** Required for large plantations (Sugar cane). Requires special permits and Detect-and-Avoid radar.

8 Review Questions & Lab Assignment

1. **Calculation:** You need to map a 50-hectare Maize farm. You have a fixed-wing drone with a 24MP camera (Sensor width 23.5mm, Focal length 16mm). You need a GSD of 3cm/pixel.
    - What altitude do you fly at?
    - If your battery lasts 45 minutes and you fly at 15 m/s, can you cover the area in one flight? (Assume 70% efficiency for turns).
2. **Physics:** Explain why a Thermal Camera requires a lens made of Germanium instead of Glass. How does this impact the cost of the drone?
3. **Design:** You are designing a "Quelea Bird Control" drone for night ops. What sensors do you need? How do you calculate the link budget for the video feed at night?

References

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